HEADLIGHT CLEANER SYSTEM

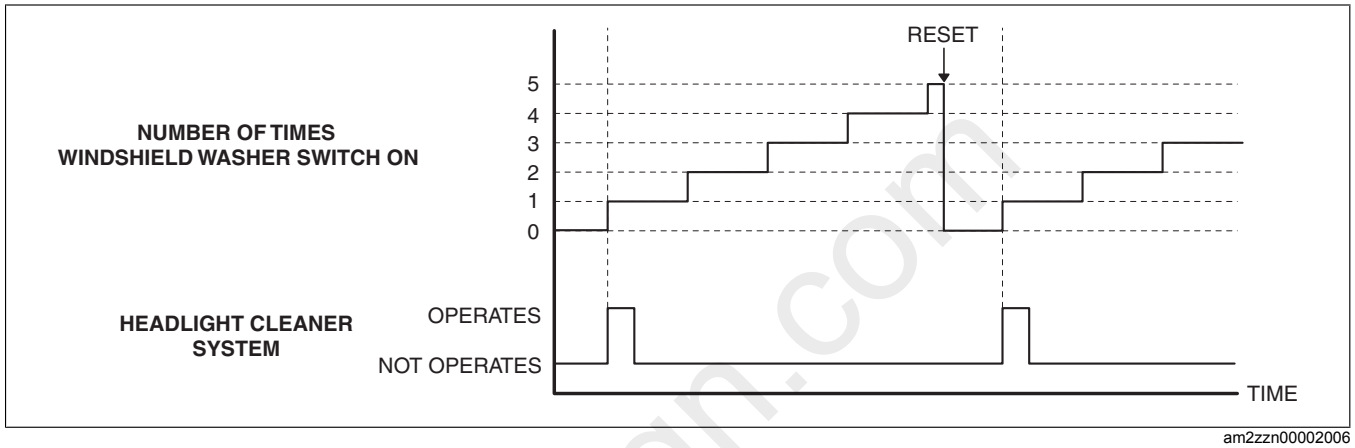
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Outline

- The headlight cleaner system injects washer fluid from the headlight cleaner nozzles equipped on the front bumper to remove dirt from the headlights.
- A pop-up type headlight cleaner has been adopted.

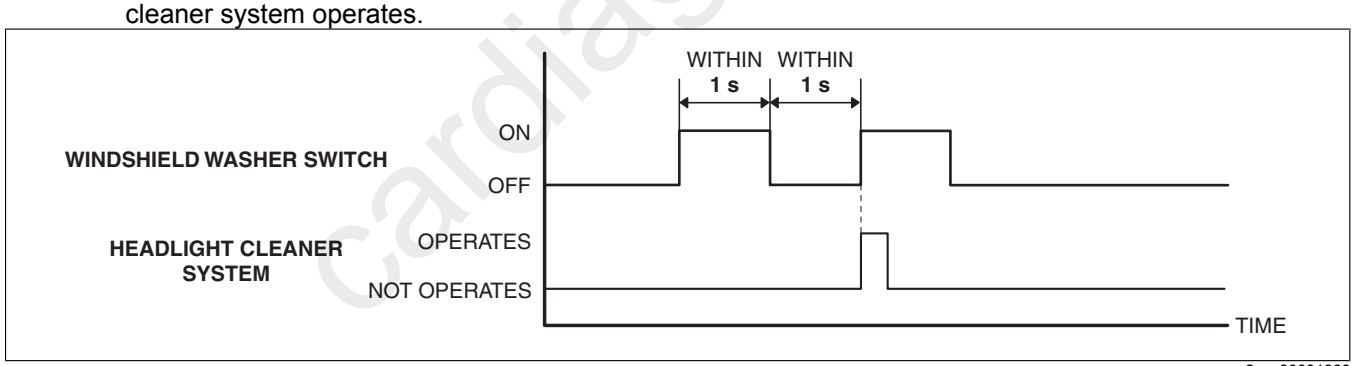
Function

- With the ignition switched ON (engine off or on) and the headlights turned on, the BCM operates the headlight cleaner system according to the windshield washer switch operation.
- The headlight cleaner actuator is activated by the pressure of the washer fluid supplied from the headlight cleaner motor, the washer fluid is injected from the headlight cleaner nozzles.
  - — Auto operation control
    - When the windshield washer switch on signal is detected the first time with the headlights turned on, the headlight cleaner system operates. Afterwards, the headlight cleaner system operates every fifth times the windshield washer switch is turned on.



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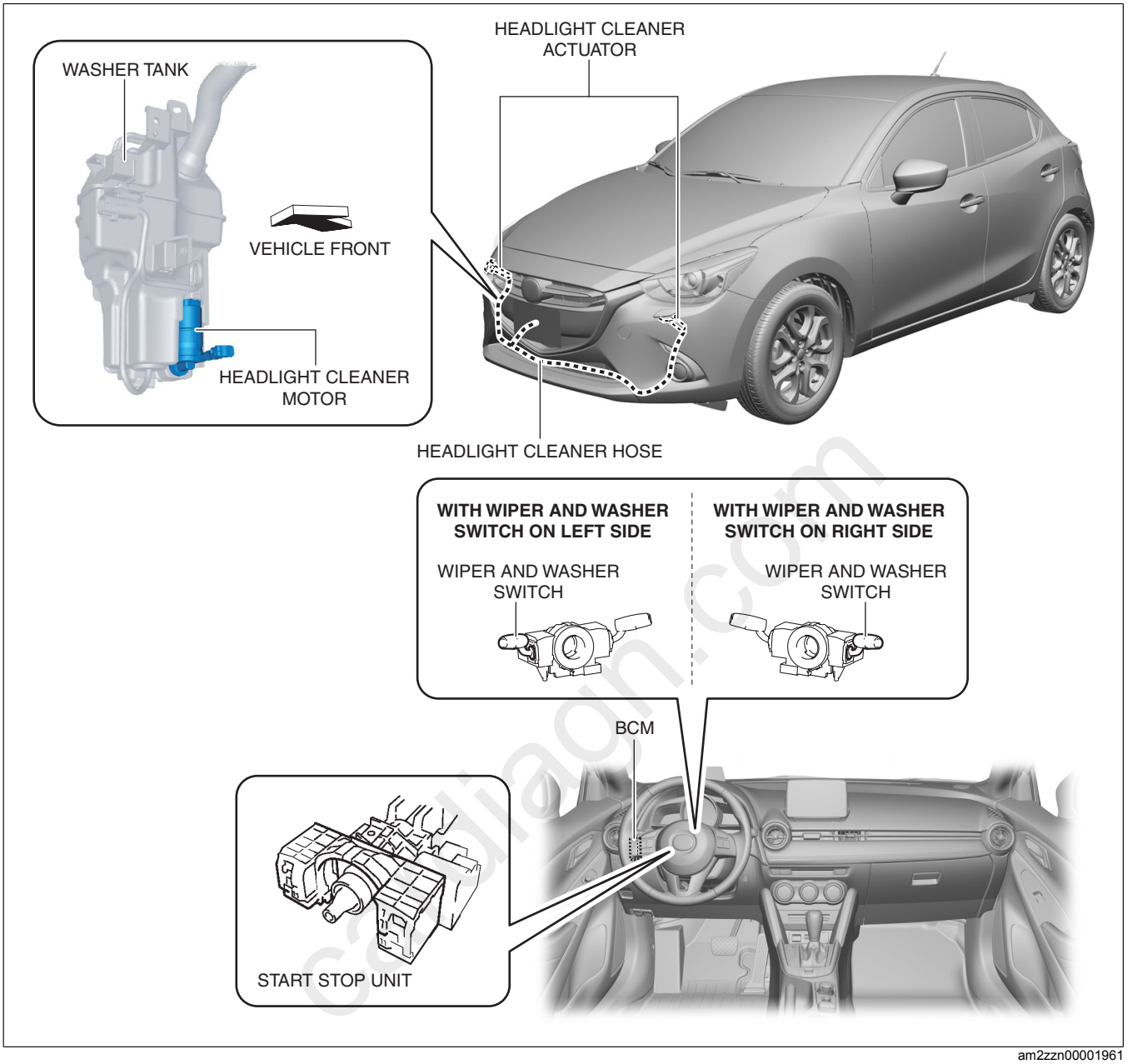
  - — Manual operation control
    - When the windshield washer switch is on two times within the range shown in the figure, the headlight cleaner system operates.



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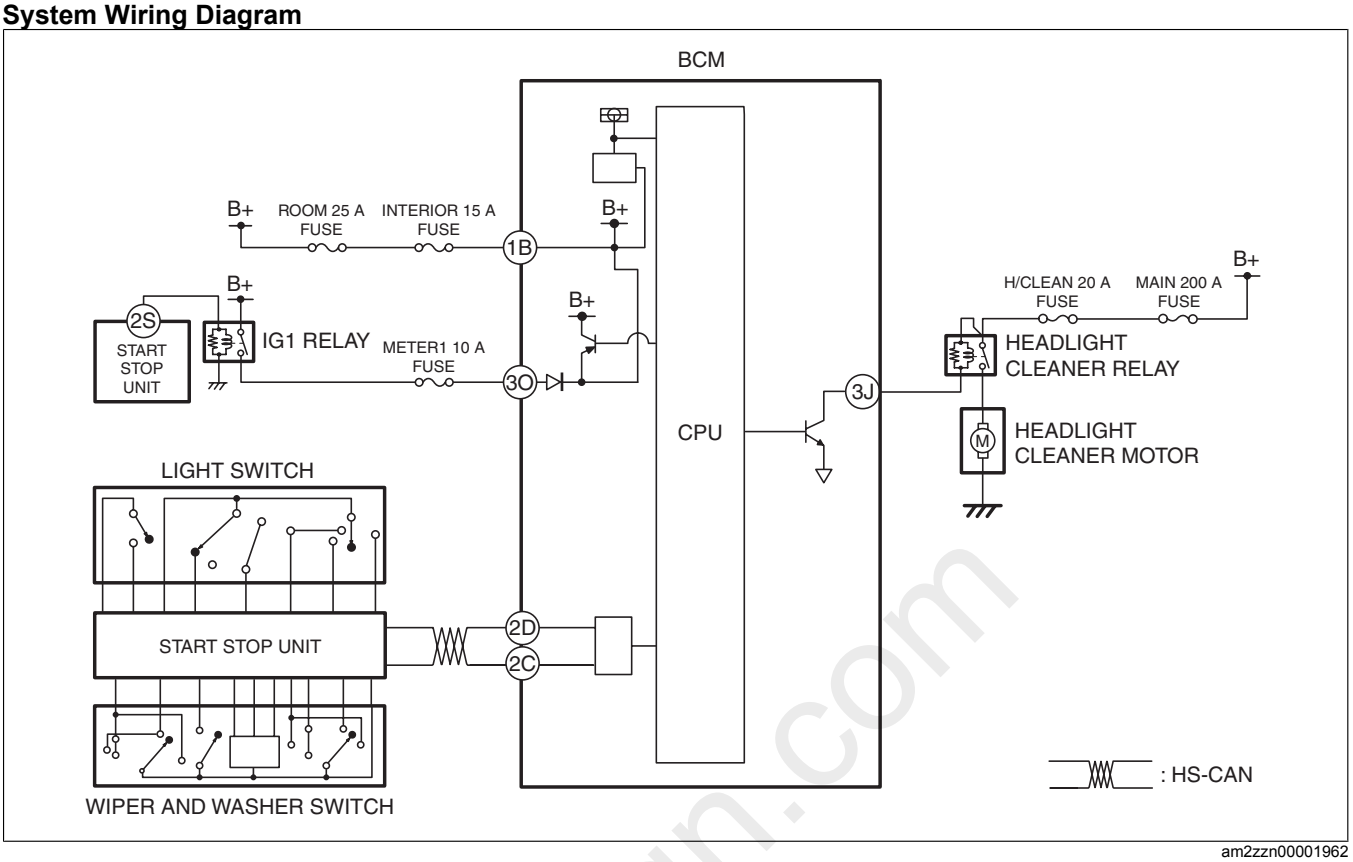
Structural View



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System Wiring Diagram



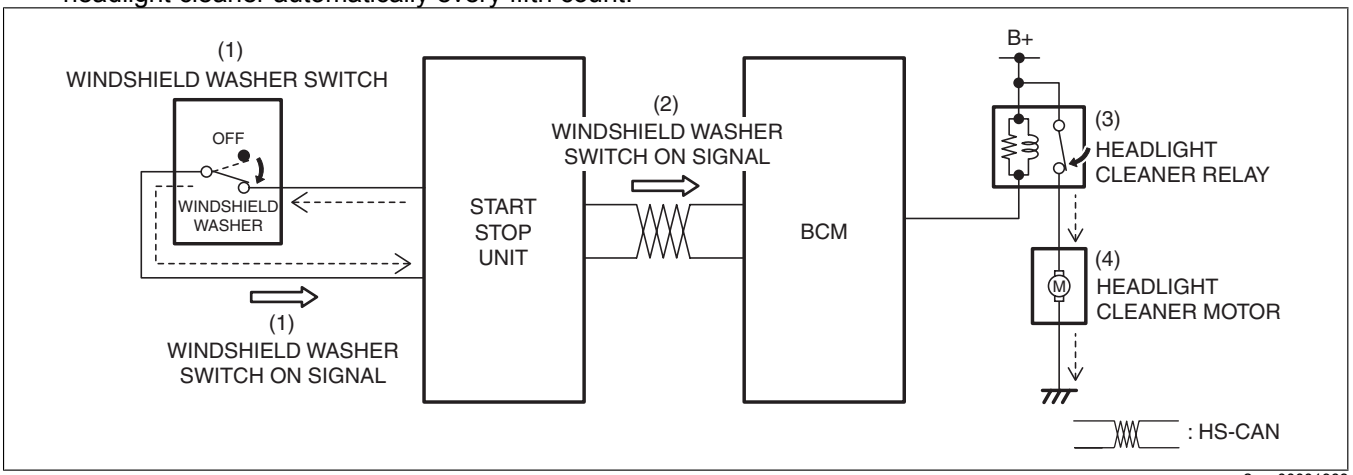
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Operation
Auto operation
Operation condition

- The headlight cleaner system operates when the following conditions are met.
  - — Ignition is switched ON (engine off or on)
  - — Headlights are turned on at LO or HI
  - — After above conditions are met, windshield washer switch is on the first time or fifth time

Operation

1. When the windshield washer switch is turned on (1), the start stop unit sends (2) a windshield washer switch on signal to the BCM via CAN communication.
2. If the BCM receives the windshield washer switch on signal 2 times within the specified time and with the operation conditions met, it turns the headlight cleaner relay on (3).
3. When the headlight cleaner relay is on, the headlight cleaner motor is activated (4) and washer fluid is injected from the headlight cleaner nozzles.
4. The BCM counts the number of times the windshield washer switch on signal is received, and operates the headlight cleaner automatically every fifth count.



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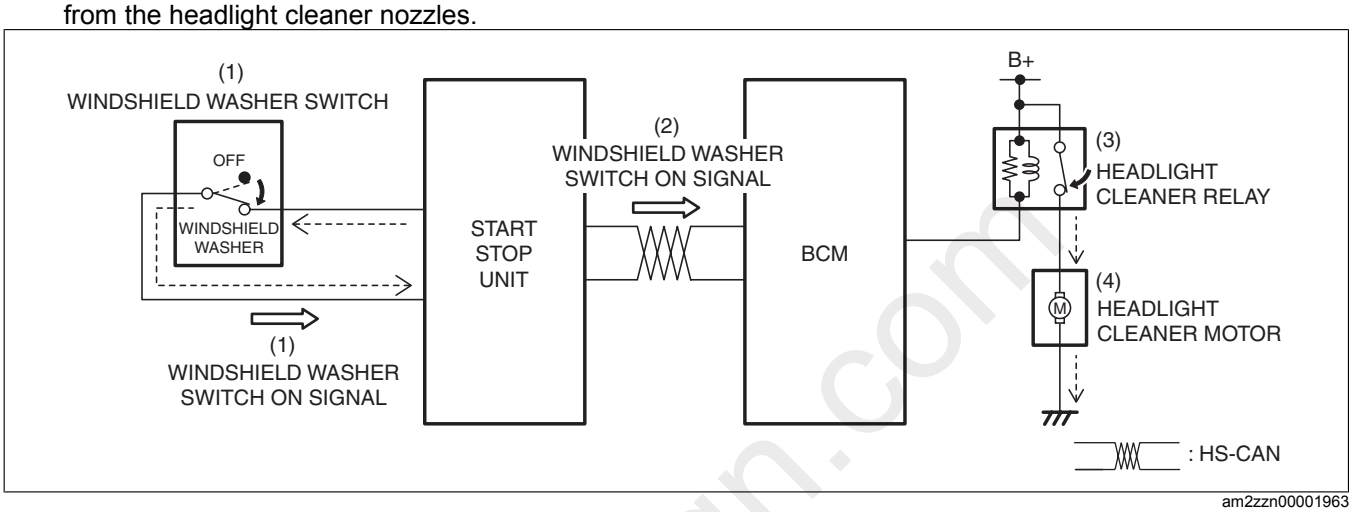
Manual operation

Operation condition

- The headlight cleaner system operates when the following conditions are met.
  - — Ignition is switched ON (engine off or on)
  - — Headlights are turned on at LO or HI
  - — After above conditions are met, windshield washer switch is on two times within a certain period of time

Operation

1. When the windshield washer switch is on (1), the start stop unit sends (2) an windshield washer switch on signal to the BCM via CAN communication.
2. When the BCM receives the windshield washer switch on signal two times within a certain period of time with the operation conditions met, it turns the headlight cleaner relay on (3).
3. When the headlight cleaner relay is on, the headlight cleaner motor is activated (4) and washer fluid is injected from the headlight cleaner nozzles.



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